No 20 - oktober - 2023

Date Jum'at

TRANSFORMASI - Z

persamaan :

$$X[z] = \sum_{-\infty}^{\infty} x[n]\, z^{-n}$$

$x[n]$: merupakan sinyal hasil sampling

n : dimulai dari 0, 1, 2, 3

A. karakteristik sinyal (Jenis)

① . causal → $x[n]$, $n \geq 0$

contoh :

$x[n] = [2 \;\; 3 \;\; 1 \;\; 4]$ → $x[0] = 2$ $x[2] = 1$

saat $n = 0, 1, 2, 3$ $x[1] = 3$ $x[3] = 4$

$X_1[z] = x[0].z^{-0} + x[1].z^{-1} + x[2].z^{-2} + x[3].z^{-3}$

$= 2 + 3.z^{-1} + z^{-2} + 4.z^{-3}$ //

② . Anti causal → $x[n]$, $n < 0$

contoh :

$x[n] = [4 \;\; 1 \;\; 2 \;\; 3]$ → $x[-3] = 4$ $x[-1] = 2$

saat $n = -3, -2, -1, 0$ $x[-2] = 1$ $x[0] = 3$

$X_2[z] = x[-3].z^{3} + x[-2].z^{2} + x[-1].z^{1} + x[0].z^{-0}$

$= 4.z^{3} + z^{2} + 2.z^{1} + 3$ //

③ . Non causal → $x[n]$, $-\infty < n < \infty$

contoh :

$x[n] = [3 \;\; 2 \;\; 2 \;\; 1 \;\; 4]$ → $x[-2] = 3$ $x[0] = 2$ $x[2] = 4$

saat $n = -2, -1, 0, 1, 2$ $x[-1] = 2$ $x[1] = 1$

$X_3[z] = x[-2].z^{2} + x[-1].z^{1} + x[0].z^{0} + x[1].z^{-1} + x[2].z^{-2}$

$= 3.z^{2} + 2.z^{1} + 2 + z^{-1} + 4z^{-2}$ //

B. Region of Convergence (ROC) → Daerah berlakunya z (cobalah ∞ dan 0

↳ langkah pengerjaan : menyubtitusikan nilai z dengan nilai - nilai ekstrim

$z\,[0, \infty, -\infty, 1]$

contoh :

1. $X_1[z] = 2 + 3z^{-1} + z^{-2} + 4z^{-3}$

saat $z = 0$ $= 2 + 3(0)^{-1} + (0)^{-2} + 4.(0)^{-3}$ tidak boleh, karena :

$= 2 + \frac{3}{0^1} + \frac{1}{0^2} + \frac{4}{0^3}$ roc untuk $X_1[z] = z \neq 0$

$z = 0 = \infty$ //

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$(\infty)^{-n} = 0$

$(0)^{-n} = \infty$

$z = \infty \rightarrow : 2 + 3z^{-1} + z^{-2} + 9z^{-3}$

$: 2 + 3(\infty)^{-1} + (\infty)^{-2} + 9.(\infty)^{-3}$

$: 2 + 0 + 0 + 0$

$z = \infty : 2$ //

$z = 1 \rightarrow : 2 + 3z^{-1} + z^{-2} + 9z^{-3}$

$: 2 + 3(1)^{-1} + (1)^{-2} + 9.(1)^{-3}$

$: 2 + 3 + 1 + 9$

$z = 1 : 10$ //

2. $X_2[z] = 9z^3 + z^2 + 2z + 3$

$z = 0 \rightarrow : 9z^3 + z^2 + 2z + 3$

$= 9(0)^3 + (0)^2 + 2(0) + 3$

$z = 0 : 3$ //

$z = \infty \rightarrow : 9z^3 + z^2 + 2z + 3$

$: 9(\infty)^3 + (\infty)^2 + 2(\infty) + 3$

$z = \infty : \infty$ //

Tidak boleh, karena : ROC (z), untuk $X_2[z]$ = $z \neq \infty$

$z = 1 \rightarrow = 9z^3 + z^2 + 2z + 3$

$= 9(1)^3 + (1)^2 + 2(1) + 3$

$= 9 + 1 + 2 + 3$

$z = 1 = 10$ //

3. $X_3[z] = 3z^2 + 2z + 2 + z^{-1} + 2z^{-2}$

$z = 0 \rightarrow : 3z^2 + 2z + 2 + z^{-1} + 9z^{-2}$

$= 3(0)^2 + 2(0) + 2 + (0)^{-1} + 9.(0)^{-2}$

$z = 0 = \infty$ //

$z = \infty \rightarrow : 3z^2 + 2z + 2 + z^{-1} + 9z^{-2}$

$= 3(\infty)^2 + 2(\infty) + 2 + (\infty)^{-1} + 9(\infty)^{-2}$

$= \infty$ //

Tidak boleh, karena : ROC (z) untuk $X_3[z]$: $z \neq 0$ dan $z \neq \infty$

$z = 1 \rightarrow : 3z^2 + 2z + 2 + z^{-1} + 9z^{-2}$

$: 3(1)^2 + 2(1) + 2 + (1)^{-1} + 9(1)^{-2}$

$= 12$ //

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c. Sinyal - sinyal yang sering digunakan

①. Sinyal impulse / delta / Dirac

persamaan :

$$x[z] = \sum_{n=-\infty}^{\infty} \delta(n) \cdot z^{-n}$$

kesimpulan :

ROC z untuk $\mathcal{Z}[\delta[n]]$ untuk semua z anggota complex

contoh :

$$x[n] = \delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{else} \end{cases}$$

$$= \sum_{n=0}^{\infty} \delta[0] \cdot z^{-0}$$

$$x[n] = 1$$

②. Sinyal tangga / step

persamaan :

$$X[z] = \mathcal{Z}[x[n]] = \mathcal{Z}[u[n]] = \sum_{n=-\infty}^{\infty} u(n) \cdot z^{-n} = \sum_{n=0}^{\infty} z^{-n}$$

$$= z^{-0} + z^{-1} + z^{-2} + z^{-3} + \dots + z^{-n}$$

$$x[z] = 1 + z^{-1} + z^{-2} + z^{-3} + \dots + z^{-n}$$

$$z^{-1}x[z] = z^{-1} + z^{-2} + z^{-3} + \dots + z^{-n}$$

$$x[z] - z^{-1}x[z] = 1$$

$$x[z](1 - z^{-1}) = 1$$

$$x[z] = \frac{1}{(1 - z^{-1})} \;//$$

maka,

$$x[z] = \frac{1}{(1 - z^{-1})} = 1 + z^{-1} + z^{-2} + z^{-3} + \dots + z^{-\infty}$$

$$x[0] = 1 + (0)^{-1} + (0)^{-2} + (0)^{-3} + \dots + (0)^{-\infty}$$

$$= 1 + \infty + \infty + \infty + \dots \infty$$

$$= \infty \;//$$

$$x[1] = 1 + (1)^{-1} + (1)^{-2} + (1)^{-3} + \dots + (1)^{-\infty}$$

$$= \infty \;//$$

$$x[\infty] = 1 + (\infty)^{-1} + (\infty)^{-2} + (\infty)^{-3} + \dots + (\infty)^{-\infty}$$

$$= 1 \;//$$

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$\frac{k}{\infty} = 0$, $\frac{k}{-\infty} = 0$

$k^{\infty} \cdot \infty \rightarrow k > 0$ } jadinya $= 0$

$k^{\infty} = 0 \rightarrow k < 0$

NOTE :

* jumlah deret bilangan :

- untuk $r > 1$ maka, $S_n = \frac{a(r^n - 1)}{r - 1}$
- untuk $r < 1$ maka, $S_n = \frac{a(1 - r^n)}{1 - r}$

S_n = jumlah bilangan untuk suku - n

a : suku pertama

n : ∞

r : rasio → perbandingan Ns dengan Bs

contoh :

$x[z] = 1 + z^{-1} + z^{-2} + z^{-3} + \ldots + z^{-\infty}$

$x[z] = 1 + (z)^{-1} + (z)^{-2} + (z)^{-3} + \ldots + (z)^{-\infty}$

$= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots + \frac{1}{\infty}$

$r = \frac{1}{2} - 1 = \frac{1}{2} \rightarrow r < 1$

$a = 1$

$n = \infty$

$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{1\left(1 - \frac{1}{2}^{\infty}\right)}{1 - \frac{1}{2}}$

$x[z] = 2$

maka, ROC (z) untuk $x[z] = \frac{1}{1 - z^{-1}}$ adalah $|z| > 1$

③. Sinyal eksponensial

$x[n] = a^n \underbrace{u[n]} = \begin{cases} a^n & n \geq 0 \\ 0, \text{ else} \end{cases}$ Deposit causal

$1, n \geq 0$ ← switch on

$0, \text{else } n < 0$

$x[z] = \mathcal{Z}[x[n]] = \mathcal{Z}(a^n u(n)) = \sum_{n=0}^{\infty} a^n z^{-n}$

$x[z] = a^0 z^{-0} + a^1 z^{-1} + a^2 z^{-2} + \ldots a^{\infty} z^{-\infty}$

$= 1 + (az^{-1})^1 + (az^{-1})^2 + \ldots (az^{-1})^{\infty}$

$az^{-1} x[z] = az^{-1} + a^2 z^{-2} + a^3 z^{-3} + a^4 z^{-4} + \ldots a^{\infty} z^{\infty} - \ldots$

maka,

$(1 - az^{-1}) x[z] = 1$

$x[z] \rightarrow (1 - az^{-1}) x[z] = \sum_{n=0}^{\infty} a^n \rightarrow \frac{1}{1-0}$, for $|a| < 1$

$x[z] = \frac{1}{1 - az^{-1}} \cdot \frac{z}{z} = \frac{z}{z - a}$

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$$x[z] = \sum_{n=n}^{\infty} a^n z^{-n}$$

✳ tabel transformasi $n \leftrightarrow z$

no.	Domain n	domain z	Roc |z|
1.	$x[n] = \delta[n] = \begin{cases} 1, n=0 \\ 0, \text{else} \end{cases}$	$x[z] = 1$	$[z]\ \forall \in \mathbb{C}$
2.	$x[n] = u[n], \begin{cases} 1, n \geq 0 \\ 0, \text{else} \end{cases}$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$\lvert z \rvert > 1$
3.	$x[n] = a^n u[n] = \begin{cases} a^n, n \geq 0 \\ 0, \text{else} \end{cases}$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$\lvert z \rvert > a$

D. karakteristik transformasi z

↳ terdapat karakter, yaitu:

① Linear

$x_1[n] \rightarrow x_1[z]$ maka, $Ax_1[n] + Bx_2[n] \rightarrow Ax_1[z] + Bx_2[z]$

$x_2[n] \rightarrow x_2[z]$

A dan B merupakan konstanta anggota bilangan kompleks dan Bilangan real.

- contoh:

1. tentukan $x[z]$ bila diketahui $x[n] = 3.\delta[n] + 4u[n]$

Jawab:

$x_1[n] = \delta[n] \rightarrow x_1[z] = 1$

$x_2[n] = u[n] \rightarrow x_2[z] = \frac{1}{1-z^{-1}}$

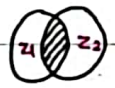
maka,

$x[z] = 3.x_1[n] + 4.x_2[n]$

$x[z] = 3.1 + 4.\frac{1}{1-z^{-1}}$

maka,

$\text{Roc}\ [z] \overset{\text{adalah}}{=} [z] = [z_1] \cap [z_2]$



$\text{Roc}\ [z_1] \overset{\text{adalah}}{=} [z_1]\ \forall \in \mathbb{C}$

$\text{Roc}\ [z_2] \overset{\text{adalah}}{=} [z_2] > 1$

maka,

Roc [z] adalah $|z| > 1$.

2. Tentukan $x[z]$ bila diketahui $x[n] = 3.u[n] + 4.2^n.u[n]$

Jawab:

$x_1[n] = u[n] \rightarrow x_1[z] = \frac{1}{1-z^{-1}}$

$x_2[n] = 2^n.u[n] \rightarrow x_2[z] = \frac{z}{z-a}$ karena $a = 2$ maka, $x_2[z] = \frac{z}{z-2}$

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maka,

$x[z] = 3.\ X_1[n] + 4.\ X_2[n]$

$= 3.\ \frac{z}{z-1} + 4.\ \frac{z}{z-2}$

maka,

poc $[z_1]$ adalah $[z_1] > 1$

poc $[z_2]$ adalah $[z_2] > 2$

} poc $[z]$ adalah $|z| > 2$ //

3. Tentukan $x(z)$ = poc $[z]$ bila diketahui $x[n] = 2.(-2)^n\ u(n) - 3(3)^n\ u(n)$

jawab :

$x_1[n] = a^n u(n) \rightarrow x_1[z] = \frac{z}{z-a}$, karena $a = -2$ maka $x[z] = \frac{z}{z+2}$

$x_2[n] = a^n u(n) \rightarrow x_2[z] = \frac{z}{z-a}$, karena $a = 3$ maka $x[z] = \frac{z}{z-3}$

maka,

$x[z] = 2.X_1[n] - 3.X_2[n]$

$= 2.\ \frac{z}{z+2} - 3.\ \frac{z}{z-3}$ //

maka,

poc $[z_1]$ adalah $[z_1] > 2$

poc $[z_2]$ adalah $[z_2] > 3$

} poc $|z|$ adalah $|z| > 3$ //

② . Time Shifting

untuk $x[z] = \mathcal{Z}[x(n)] \rightarrow \mathcal{Z}[x[n-k]] = z^{-k} X[z]$

poc $[z]$ = $[z]$ pada $x[z]$

- contoh :

1. tentukan $x[z]$ dan poc $[z]$ bila diketahui $x[n] = 5.\ u(n-2)$

jawab :

$x_1[n] = u[n] \rightarrow x_1[z] = \frac{1}{1-z^{-1}}$

$x_2[n] = u(n-2) \rightarrow x_2[z] = z^{-2}.\ \frac{1}{1-z^{-1}}$

maka,

$x[n] = 5.\ u(n-2) \rightarrow x[z] = 5.\ z^{-2}\ \frac{1}{1-z^{-1}}$ //

maka,

poc = $|z| > 1$ //

Diliat dari tabel poc yang awalnya diguratan

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2. $x[n] = 3\delta[n-1] + 2 \cdot u[n-5]$

Jawab :

$x_1[n] = \delta[n] \longrightarrow X_1[z] = 1$

$x_2[n] = \delta[n-1] \longrightarrow X_2[z] = z^{-1} \cdot 1$

$x_3[n] = u[n] \longrightarrow X_3[z] = \frac{1}{1-z^{-1}}$

$x_4[n] = u[n-5] \longrightarrow X_4[z] = z^{-5} \cdot \frac{1}{1-z^{-1}}$

maka,

poc $[z_1]$ adalah $|z| \forall \in \mathbb{C}$

poc $[z_2]$ adalah $|z| > 1$

maka,

poc $[z]$ adalah $|z| > 1$ //

maka,

$x[n] = 3\delta[n-1] + 2 \cdot u[n-5] \longrightarrow X[z] = 3z^{-1} + 2 \cdot z^{-5} \cdot \frac{1}{1-z^{-1}} = \frac{3}{z} + 2z^{-5} \cdot \frac{z}{z-1}$ //

$= \frac{3}{z} + \frac{2z^{-4}}{z-1}$

$= \frac{3}{z} + \frac{2}{(z-1)(z^4)}$ //

2. Tentukan $x[z]$ dan poc $[z]$ bila diketahui

$x[n] = 3 \cdot (2)^{n-2} u[n-2] - 5(-3)^{n-4} u[n-4]$

Jawab :

$x_1[n] = a^n u[n] \longrightarrow X_1[z] = \frac{1}{1-az^{-1}}$ $a=2$ $X_1[z] = \frac{1}{1-2z^{-1}}$

$x_1[n] = a^{n-2} u[n-2] \longrightarrow X_1[z] = z^{-2} \cdot \frac{1}{1-2 \cdot z^{-1}}$

$x_2[n] = a^n u[n] \longrightarrow X_2[z] = \frac{1}{1-az^{-1}}$ $a=-3$ $X_1[z] = \frac{1}{1+3 \cdot z^{-1}}$

$x_2[n] = a^{n-4} u[n-4] \longrightarrow X_2[z] = z^{-4} \cdot \frac{1}{1+3 \cdot z^{-1}}$

maka,

$x[n] = 3 \cdot z^{-2} \cdot \frac{1}{1-2 \cdot z^{-1}} - 5 \cdot z^{-4} \frac{1}{1+3 \cdot z^{-1}}$ //

maka,

poc $[z_1]$ adalah $|z| > 2$

poc $[z_2]$ adalah $|z_2| > 3$

maka,

poc $[z]$ adalah $|z| > 3$ //

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$\mathcal{Z}$: transformasi

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⑤. Stalling : tabel no 3

untuk $x[z] : \mathcal{Z}[x(n)] \rightarrow \mathcal{Z}[k^n x[n]] : x\left[\frac{z}{k}\right]$

$x[n] : u(n) \rightarrow x[z] : \frac{1}{1-z^{-1}}$

$x[n] : a^n u(n) \rightarrow x[z] : \frac{1}{1-\left(\frac{z}{a}\right)^{-1}} : \frac{1}{1-\frac{z^{-1}}{a^{-1}}} : \frac{1}{1-az^{-1}}$

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④. konvolusi dalam domain n = perkalian dalam domain z

$$Z[x_1[n] * x_2[n]] = X_1[z] \cdot X_2[z]$$

$$Z[x[n]] = \sum_{n=-\infty}^{n} x[n] z^{-n}$$

⇒ cara :

Tentukan $Y[z] = Z[x_1[n] * x_2[n]]$

1). konvolusi $y[n] = x_1[n] * x_2[n]$, lalu $Y[z] = Z[y_1[n]]$

2). Ubah dulu $Y_1[z] = Z[x_1[n]]$, $Y_2[z] = Z[x_2[n]]$ } $Y[z] = Y_1[z] \cdot Y_2[z]$

$Y[n] = Z^{-1}[Y[z]]$

contoh :

1. Diketahui $x_1[n] = [1\ \underset{0}{3}\ \underset{1}{2}]$

$x_2[n] = [\underset{0}{2}\ \underset{1}{1}]$

Jawab cara 1 :

$x_1[n] = [\underset{0}{1}\ \underset{1}{3}\ \underset{2}{2}]$

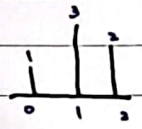
$x_2[n] = [\underset{0}{2}\ \underset{1}{1}]$

$M = N + K - 1$

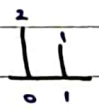
$= 3 + 2 - 1$

$= 4$ proses

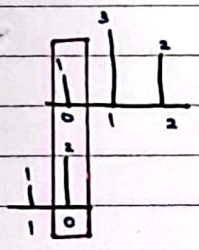
$x_1[n]$ =



$x_2[n]$ =

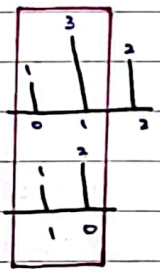


$Y(0)$



$Y(0) = 2.1 = 2$

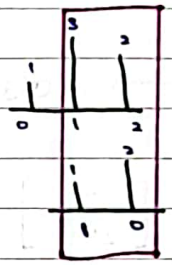
$Y(1)$



$Y(1) = (1.1) + (3.2)$

$Y(1) = 7$

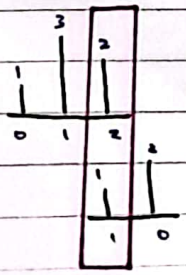
$Y(2)$



$Y(2) = (3.1) + (2.2)$

$Y(2) = 7$

$Y(3)$



$Y(3) = 2.1 = 2$

$Y(n) = [2\ \underset{0}{7}\ \underset{1}{7}\ \underset{2}{2}]$

$$y(z) = \sum_{n=0}^{3} y(n) \cdot z^{-n}$$

$= y(0).z^{-0} + y(1).z^{-1} + y(2).z^{-2} + y(3).z^{-3}$

$Y(z) = 2 + 7z^{-1} + 7z^{-2} + 2z^{-3}$ //

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Jawab cara 2 :

$$Y[z] = [x_1[n] * x_2[n]]$$

$$Y_1[z] = \sum_{n=0}^{2} x[n] \cdot z^{-n} = x(0) \cdot z^{-0} + x[1] \cdot z^{-1} + x(2) \cdot z^{-2}$$
$$= 1 \cdot z^{-0} + 3 \cdot z^{-1} + 2 \cdot z^{-2}$$

$$Y_2[z] = \sum_{n=0}^{1} x[n] \cdot z^{-n} = x[0] \cdot z^{-0} + x[1] \cdot z^{-1}$$
$$= 2 \cdot z^{-0} + 1 \cdot z^{-1}$$

$$Y[z] = Y_1[z] \cdot Y_2[z]$$
$$= (z^{-0} + 3z^{-1} + 2z^{-2}) \cdot (2z^{-0} + z^{-1})$$
$$= (1 + 3z^{-1} + 2z^{-2}) \cdot (2 + z^{-1})$$
$$= 2 + z^{-1} + 6z^{-1} + 3z^{-2} + 4z^{-2} + 2z^{-3}$$
$$Y[z] = 2 + 7z^{-1} + 7z^{-2} + 2z^{-3} //$$

⑤. Diferensiasi dalam daerah z

untuk $x[z] = \mathcal{Z}[x[n]]$

$$\therefore \mathcal{Z}[n \cdot x[n]] = -z \left[\frac{d}{dz} \cdot x[z] \right]$$

$$ROC[z] = [z] \text{ in } x[z]$$

contoh :

1. Bila diketahui $x[n] = n \cdot u[n]$. tentukan $x[z]$

Jawab :

$$x_1[n] = u[n] \longrightarrow x_1[z] = \frac{z}{z-1}$$

$$\frac{d}{dz}[x_1[z]] = \frac{d}{dz}\left[\frac{z}{z-1}\right] = \frac{u'v - uv'}{v^2} \qquad u = z \quad u' = 1$$
$$v = z-1 \quad v' = 1$$
$$= \frac{1(z-1) \cdot z \cdot 1}{(z-1)^2} \cdot \frac{-1}{(z-1)^2}$$

maka,

$$x[z] = \mathcal{Z}[n \cdot x[n]] = -z \left[\frac{-1}{(z-1)^2} \right] = \frac{z}{(z-1)^2} //$$

$$ROC\ |z| > 1 //$$

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E. Definisi sinyal dan sistem dalam domain z

- Ekspresi sinyal dalam domain n : $x[n] = [\ldots\ 2\ 1\ 3\ \ldots]$
- Ekspresi sistem atau filter dalam domain n : $h[n] = [2\ 3\ 1]$

$x[z] = \ldots 2z^{-k} + 1z^{-(k+1)} + 3z^{-(k+2)} \ldots$ n [suku banyak]

$h[z] = 2z^{0} + 3z^{-1} + z^{-2}$

- Proses penyaringan :

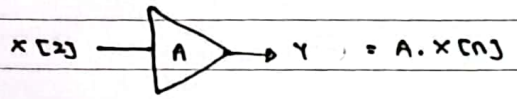
$y[k] = x[n] * h[m]$ → konvolusi

$h[m] * x[n]$ → Bersifat komutatif

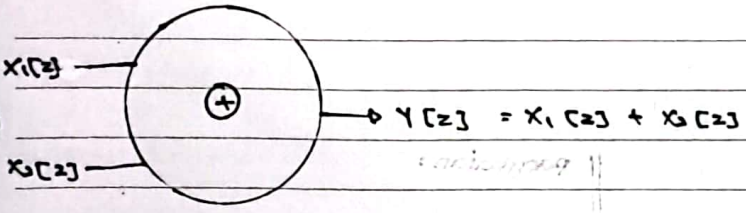
Sinyal		Sistem		
$x[n]$	→	$h[m]$	→	$y[k]$
$h[m]$	→	$x[n]$	→	$y[k]$ perhitungan boleh, realisasi gabolch

F. Rangkaian filter digital dalam domain z

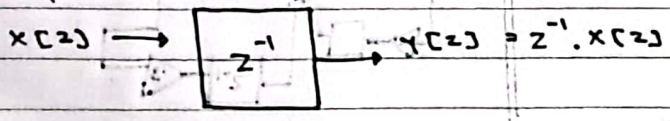
① komponen pengali



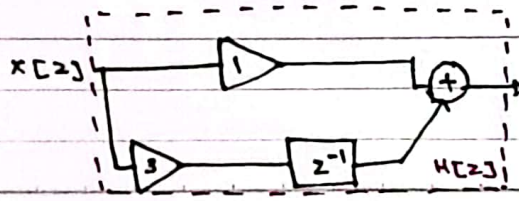
② komponen penjumlah



③ komponen penunda



- contoh soal : Buat rangkaian filter $H[z] = 1 + 3z^{-1}$



$Y[z] = 1 + 3z^{-1}$

$H[z] = \frac{Y[z]}{X[z]} \rightarrow Y[z] = X[z]\,h[z]$

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- Sintesis → proses membuat rangkaian filter dari persamaan filter H[z] yang diberikan.

$X[z] \rightarrow \boxed{H[z]} \rightarrow Y[z]$

$Y[z] = H[z] \cdot X[z]$; maka $H[z] = \frac{Y[z]}{X[z]}$

- untuk membuat rangkaiannya, harus dicari persamaan untuk Y[z]

contoh :

1. Buat rangkaian untuk $H[z] = 1 + 2z^{-1}$

Jawab :

$H[z] = \frac{Y[z]}{X[z]} = 1 + z^{-1}$

$Y[z] = (1 + z^{-1}) X[z]$

$Y[z] = X[z] + z^{-1} X[z]$

maka, rangkaiannya :



2. Buat rangkaian untuk $H[z] = \frac{1 - 2z^{-1}}{1 + 3z^{-1}}$

Jawab :

$H[z] = \frac{Y[z]}{X[z]} = \frac{1 - 2z^{-1}}{1 + 3z^{-1}}$

$Y[z] = \frac{1 - 2z^{-1}}{1 + 3z^{-1}} X[z] \longrightarrow = \frac{(1 - 2z^{-1}) X[z]}{1 + 3z^{-1}}$

$(1 + 3z^{-1}) Y[z] = (1 - 2z^{-1}) X[z]$

$Y[z] + 3z^{-1} Y[z] = X[z] - 2z^{-1} X[z]$

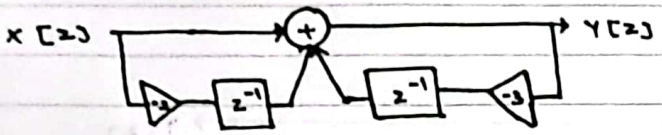
$Y[z] = X[z] - 2z^{-1} X[z] - 3z^{-1} Y[z]$

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Rangkaian :



3. Buat rangkaian untuk filter $H[z] = \frac{1-2z^{-1}+2z^{-2}}{1+z^{-2}}$

Jawab :

$$H[z] = \frac{Y[z]}{X[z]} = \frac{1-2z^{-1}+2z^{-2}}{1+z^{-2}}$$

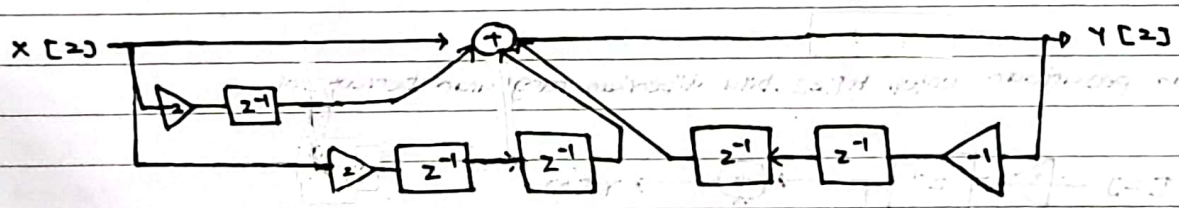
$$Y[z] = \frac{1-2z^{-1}+2z^{-2}}{1+z^{-2}} X[z] \longrightarrow = \frac{((1-2z^{-1})+(2z^{-2}))X[z]}{1+z^{-2}}$$

$$(1+z^{-2})Y[z] = (1-2z^{-1}+2z^{-2})X[z]$$

$$Y[z] + z^{-2}Y[z] = X[z] - 2z^{-1}X[z] + 2z^{-2}X[z]$$

$$Y[z] = X[z] - 2z^{-1}X[z] + 2z^{-2}X[z] - z^{-2}Y[z]$$

Rangkaian



4. Buat rangkaian untuk filter $H[z] = \frac{z^2+2z+1}{z^2-3z-1}$

Jawab :

$$H[z] = \frac{Y[z]}{X[z]} = \frac{z^2+2z+1}{z^2-3z-1}$$

$$Y[z] = \frac{z^2+2z+1}{z^2-3z-1} \cdot X[z]$$

$$(z^2-3z-1)Y[z] = X[z](z^2+2z+1)$$

$$z^2Y[z] - 3zY[z] - Y[z] = z^2X[z] + 2zX[z] + X[z]$$

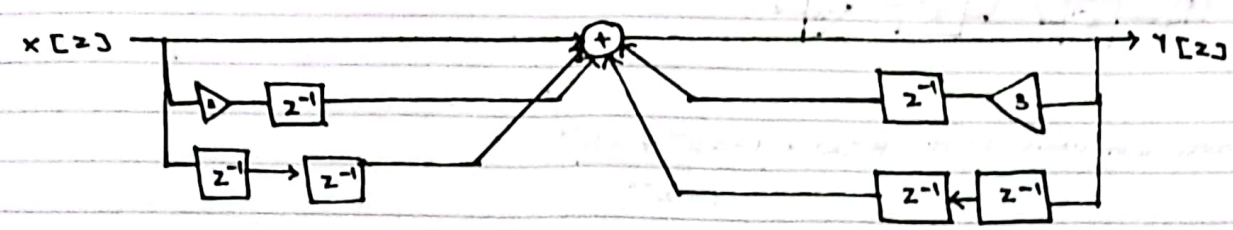
$$z^2Y[z] = z^2X[z] + 2zX[z] + X[z] + 3zY[z] + Y[z]$$

$$Y[z] = \frac{z^2X[z] + 2zX[z] + X[z] + 3zY[z] + Y[z]}{z^2}$$

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$Y[z] = X[z] + 2z^{-1} X[z] + z^{-2} X[z] + 3z^{-1} Y[z] + z^{-2} Y[z]$

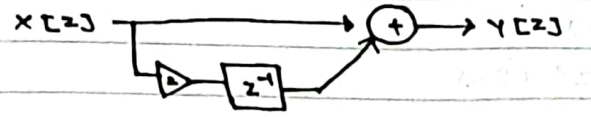
Rangkaian :



• Analisa → proses merumuskan persamaan H[z] dari rangkaian filternya

contoh :

1. Tentukan persamaan untuk H[z] bila diberikan rangkaian filter berikut ini



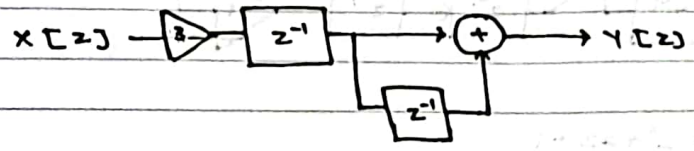
Jawab :

$Y[z] = X[z] + 2z^{-1} X[z]$

$Y[z] = (1 + 2z^{-1}) X[z]$

maka, $H[z] = \frac{Y[z]}{X[z]} = 1 + 2z^{-1}$ //

2. Tentukan persamaan untuk H[z] bila diberikan rangkaian berikut ini



Jawab :

$Y[z] = 3z^{-1} X[z] + 3z^{-2} X[z]$

$Y[z] = (3z^{-1} + 3z^{-2}) X[z]$

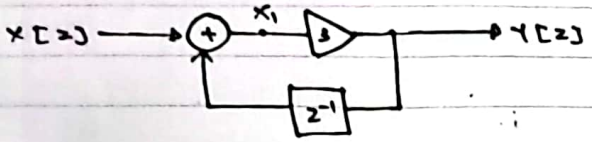
maka, $H[z] = \frac{Y[z]}{X[z]} = 3z^{-1} + 3z^{-2}$ //

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3. Tentukan H[z] pada rangkaian berikut



Jawab :

$Y[z] = 3X_1$

$Y[z] = 3(X[z] + z^{-1}Y[z])$

$Y[z] = 3X[z] + 3z^{-1}Y[z]$

$Y[z] - 3z^{-1}Y[z] = 3X[z]$

$(1 - 3z^{-1})Y[z] = 3X[z]$

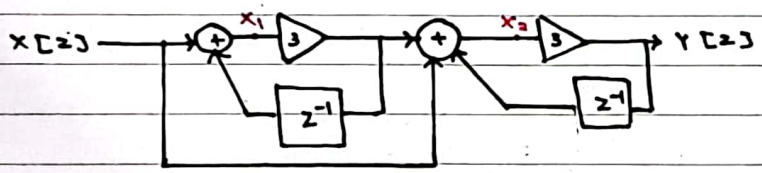
maka,

$H[z] = \frac{Y[z]}{X[z]} = \frac{3}{1-3z^{-1}}$ //

dibalik

$\frac{Y[z]}{X[z]} = \frac{1}{1-3z^{-1}} \cdot 3$

4. Tentukan persamaan H[z] pada rangkaian berikut



Jawab :

$X_1 = X[z] + 3z^{-1}X_1$

$X_1 - 3z^{-1}X_1 = X[z]$

$(1 - 3z^{-1})X_1 = X[z]$

$X_1 = \frac{X[z]}{1-3z^{-1}}$

$X_2 = X[z] + 3X_1 + 3z^{-1}X_2$

$X_2 - 3z^{-1}X_2 = X[z] + 3X_1$

$(1 - 3z^{-1})X_2 = X[z] + 3X_1$

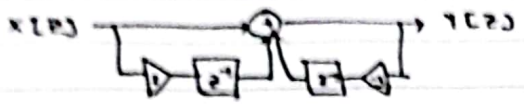
$(1 - 3z^{-1})X_2 = X[z] + 3\left(\frac{X[z]}{1-3z^{-1}}\right)$

$(1 - 3z^{-1})X_2 = X[z] + \frac{3X[z]}{1-3z^{-1}} : 1 - 3z^{-1}$

$X_2 = \frac{X[z]}{1-3z^{-1}} + 3X[z]$

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* sebutkan persamaan $h[n]$ dan rangkaian berikut :



jawab :

$y[n] = x[n] + 2x[n-1] - 3y[n-1]$

$y[n] + (3y[n-1]) = x[n] + 2x[n-1]$

$y[n](1+3z^{-1}) = x[n](1+2z^{-1})$

maka, (dapatlah)

$H[z] = \frac{y[n]}{x[n]} = \frac{1+2z^{-1}}{1+3z^{-1}}$

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6. Persamaan Differensial

$x[z] \rightarrow \boxed{H[z]} \rightarrow Y[z]$

$H[z] = \frac{Y[z]}{x[z]} \rightarrow Y[z] = H[z] . x[z]$

- persamaan Differensial dalam domain z

untuk mencari persamaan differensial dalam domain -n digunakan property time shifting di dalam domain -n sbb :

$\mathcal{Z}[x[n]] = x[z] \rightarrow \therefore \mathcal{Z}[x[n-k]] = z^{-k} x[z]$

*contoh soal :

1. Diketahui $H[z] = 1 + 2z^{-1}$ finite impulse respons (FIR) filter

mata tentukan :

y. Turunkan persamaan differensial untuk $y[n]$

Jawab :

$H[z] = \frac{Y[z]}{x[z]} = 1 + 2z^{-1}$

$Y[z] = (1+2z^{-1}) . x[z] = x[z] + 2z^{-1} x[z]$

maka, persamaan differensial dalam domain -n nya adalah

$y[n] = x[n] + 2x[n-1]$ //

catatan $\mathcal{Z}^{-1}[z^{-k} x[z]] = x[n-k]$

b). Bila $x[n] = [2, 1]$, hitung $y[n]$, $n = 0, 1, 2, 3$

Jawab :

$n=0 \rightarrow y[0] = x[0] + 2x[-1] = 2$

$n=1 \rightarrow y[1] = x[1] + 2x[0] = 1 + 2.2 = 5$

$n=2 \rightarrow y[2] = x[2] + 2x[1] = 2.1 = 2$

$n=3 \rightarrow y[3] = x[3] + 2x[2] = 0$

$\vdots$

$n \rightarrow y[n] = 0$

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2. Diketahui $H[z] = \frac{1+2z^{-1}-8z^{-2}}{1+3z^{-1}-z^{-2}}$ Infinite Impulse Response (IIR) filter

a). Tentukan persamaan differensial untuk $y[n]$

Jawab:

$$H[z] = \frac{Y[z]}{X[z]} = \frac{1+2z^{-1}-8z^{-2}}{1+3z^{-1}-z^{-2}}$$

$$Y[z] = \frac{1+2z^{-1}-8z^{-2}}{1+3z^{-1}-z^{-2}} \cdot X[z]$$

$$Y[z] = \frac{X[z] + 2z^{-1}X[z] - 8z^{-2}X[z]}{1+3z^{-1}-z^{-2}}$$

$$Y[z](1+3z^{-1}-z^{-2}) = X[z] + 2z^{-1}X[z] - 8z^{-2}X[z]$$

$$Y[z] + 3z^{-1}Y[z] - z^{-2}Y[z] = X[z] + 2z^{-1}X[z] - 8z^{-2}X[z]$$

$$Y[z] = X[z] + 2z^{-1}X[z] - 8z^{-2}X[z] - 3z^{-1}Y[z] + z^{-2}Y[z]$$

$$y[n] = x[n] + 2x[n-1] - 8x[n-2] - 3y[n-1] + y[n-2] \quad //$$

b). Jika $x[n] = [2 \;\; 1 \;\; 3 \;\; 1 \;\; 2]$, hitung $y[n] = 0, 1, 2, 3, 4, 5$

$n=0 \rightarrow y[0] = x[0] + 2x[-1] - 8x[-2] - 3y[-1] + y[-2] = 2 + 2 - 0 - 0 + 0 = 4$

$n=1 \rightarrow y[1] = x[1] + 2x[0] - 8x[-1] - 3y[0] + y[-1] = 1 + 4 - 8 - (-3) + 0 = 0$

$n=2 \rightarrow y[2] = x[2] + 2x[1] - 8x[0] - 3y[1] + y[0] = 3 + 2 - 16 - 0 + 4 = -7$

$n=3 \rightarrow y[3] = x[3] + 2x[2] - 8x[1] - 3y[2] + y[1] = 1 + 6 - 8 + 21 + 0 = 20$

$n=4 \rightarrow y[4] = x[4] + 2x[3] - 8x[2] - 3y[3] + y[2] = 2 + 2 - 24 - 60 + (-7) = -87$

3. Diketahui $H[z] = \frac{2z^2+3z+1}{z^2+1}$ Tentukan $y[n]$

Jawab:

$$H[z] = \frac{2z^2+3z+1}{z^2+1} \cdot \frac{z^{-2}}{z^{-2}} = \frac{2z^0+3z^{-1}+z^{-2}}{z^0+z^{-2}} = \frac{2+3z^{-1}+z^{-2}}{1+z^{-2}}$$

$$Y[z] = \frac{2+3z^{-1}+z^{-2}}{1+z^{-2}} \cdot X[z] = \frac{2X[z] + 3z^{-1}X[z] + z^{-2}X[z]}{1+z^{-2}}$$

$$Y[z](1+z^{-2}) = 2X[z] + 3z^{-1}X[z] + z^{-2}X[z]$$

$$Y[z] + z^{-2}Y[z] = 2X[z] + 3z^{-1}X[z] + z^{-2}X[z]$$

$$Y[z] = 2X[z] + 3z^{-1}X[z] + z^{-2}X[z] - z^{-2}Y[z]$$

$$y[n] = 2x[n] + 3x[n-1] + x[n-2] - y[n-2] \quad //$$

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1. Diketahui $H[z] = \frac{1}{1+z^{-1}} + \frac{1}{1-z^{-1}}$ tentukan $y[n]$

jawab :

$$H[z] = \frac{Y[z]}{X[z]} : \frac{1}{1+z^{-1}} + \frac{1}{1-z^{-1}}$$

$$Y[z] = \left(\frac{1}{1+z^{-1}} + \frac{1}{1-z^{-1}} \right) . X[z]$$

$$= \frac{(1-z^{-1}) + (1+z^{-1})}{(1+z^{-1})(1-z^{-1})} . X[z]$$

$$= \frac{1+1}{(1+z^{-1})(1-z^{-1})} . X[z]$$

$$= \frac{2}{1-z^{-1}+z^{-1}-z^{-2}} = \frac{2}{1-z^{-2}} . X[z]$$

$$Y[z] = \frac{2X[z]}{1-z^{-2}}$$

$$(1-z^{-2})(Y[z]) = 2X[z]$$

$$Y[z] - z^{-2}Y[z] = 2X[z]$$

$$Y[z] = 2X[z] + z^{-2}Y[z]$$

$$Y[n] = 2X[n] + Y[n-2]$$ //

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